Pass-Through, Welfare, and Incidence under Product Returns in Perfect Competition

Draft notes

July 26, 2026

1 Introduction

About one in five products sold online is returned. Pass-through, tax incidence, and the efficiency cost of taxation are well understood when purchase settles who keeps the good. In that classical setting, the demand and supply curves are sufficient for consumer surplus, producer surplus, and deadweight loss. With returns, allocation is decided in two stages. Buyers first choose whether to purchase, then whether to keep or return after receiving the item. Never buying and buying then returning leave the same final allocation—the consumer does not keep the good—yet their welfare costs differ. A returned purchase incurs shipping, hassle, and handling that a non-purchase avoids, so demand and supply alone omit an economically relevant margin.

What objects then replace the classical demand and supply curves, and how misleading is an analysis that ignores returns? We study these questions under perfect competition with product returns and seller handling costs. We characterize pass-through, incidence, and welfare losses with returns, compare them to the no-return benchmark, and identify where omitting the return margin changes conclusions.

Even with a return margin, surplus continues to rest on a small set of market aggregates. What changes is which aggregates matter and at which prices they are evaluated. Consumer surplus depends only on two aggregates, gross demand and the return rate. Heterogeneity in tastes, match values, and return costs enters welfare solely through these aggregates. Producer surplus retains the usual geometry—area to the left of the supply curve—but is evaluated at net revenue, list price minus expected refunds and handling, rather than at list price.

Because consumers pay the list price while firms respond to net revenue, pass-through from cost shocks into list prices is governed by how net revenue moves with the list price. The response of net revenue to the list price reflects two forces. First, a higher price directly raises revenue on kept units. Second, the return rate itself may change with price, which alters the expected refund and handling costs the seller incurs. Whether the return rate rises or falls depends on a behavioral channel—higher prices make consumers more likely to return—and a selection channel—price changes alter the composition of buyers. When the behavioral channel dominates, a higher list price actually lowers net revenue, making supply slope downward in the list price and setting the stage for the surprising pass-through results below.

A related and equally important departure concerns tax incidence. In the standard model, it does not matter which party remits a tax. With returns, this neutrality fails because the refund price differs by remittance regime: under seller remittance, the tax-inclusive checkout price is refunded; under buyer remittance, only the pre-tax list price is. Since the return decision depends on the refund price, the two regimes generate different effective revenues and equilibria. These two departures—the behavioral response of returns to price and the failure of tax neutrality—are the central forces that shape the pass-through results below.

We show that pass-through of a one-dollar production-cost shock equals that of a seller-remitted unit tax, but differs under buyer remittance. The ratio of handling to production-cost pass-through equals the aggregate return rate. When the behavioral response of returns to price is strong enough, buyer- and seller-remitted pass-through carry opposite signs, a sharp reversal of the classical neutrality intuition. Which regime yields a negative pass-through depends on the relative magnitudes of the supply and demand elasticities. Intuitively, negative pass-through arises when the list-price supply curve slopes downward: a higher price reduces net revenue and hence quantity supplied. In this region, a cost increase creates excess demand, and the equilibrium price must fall to restore market clearing. We also provide a simple diagnostic, from the return-rate elasticity and the ratio of handling to list price, for when the classical no-return pass-through formula over- or underestimates the correct returns-adjusted pass-through.

A striking result emerges from the incidence analysis: despite pass-through of a cost shock potentially being negative, the incidence ratio is always positive and identical across all four shocks—production cost, handling cost, and both seller- and buyer-remitted taxes. This uniformity implies that consumer surplus, producer surplus, and welfare always move in the same direction, similar to the classical model without returns. Nevertheless, the welfare implications can be very different and hinge critically on the sign of pass-through. When cost pass-through is negative, production cost, handling cost, and seller tax shocks increase both surpluses, yielding a Pareto improvement. For buyer-remitted taxes when pass-through is negative, both surpluses unambiguously decline. When pass-through is positive, all four shocks reduce both surpluses. This stands in sharp contrast to the classical no-return benchmark, where pass-through is always positive and such welfare-improving outcomes are impossible.

2 Setting

We begin with the market primitives: purchase and return behavior on the consumer side, and competitive supply with return handling on the firm side. A homogeneous product is sold at list price p. Consumers indexed by i differ in return hassle κ_i, the disutility from returning the product, and in match distribution G_i over post-purchase match quality u. After purchase, type i draws $u \sim G_i$ and keeps the item whenever $u - p \geq -\kappa_i$, earning surplus $u - p$, and returns it otherwise, earning $-\kappa_i$.

Buying at price p yields expected surplus

$$cs_i(p) = \mathbb{E}_{u\sim G_i}\big[\max(u - p,\, -\kappa_i)\big].$$

On the purchase margin, all consumer-side heterogeneity is summarized by willingness to pay v_i, defined by $cs_i(v_i) = 0$, so type i buys if and only if $p \leq v_i$. Across types, willingness to pay has distribution $F(v) = \Pr(v_i \leq v)$ with density f, so gross demand is $D(p) = 1 - F(p)$. Of the units sold at price p, a share $r(p)$ are returned.

To understand how $r(p)$ changes with price, write type i's return probability as $\rho_i(p) = \Pr(u < p - \kappa_i)$. Willingness to pay is sufficient for purchase but not for returns: types with the same v_i may still differ in (κ_i, G_i). Averaging within WTP cells,

$$\rho(v, p) \equiv \mathbb{E}\big[\rho_i(p) \bigm| v_i = v\big],$$

so the aggregate return rate is the truncated mean

$$r(p) = \mathbb{E}\big[\rho(v_i, p) \bigm| v_i \geq p\big].$$

Differentiating yields

$$r'(p) = \mathbb{E}\big[\partial\rho(v_i,p)/\partial p \bigm| v_i \geq p\big] + \frac{f(p)}{D(p)}\big(r(p) - \rho(p,p)\big),$$

where $\rho(p,p)$ is the average return probability among types at the purchase margin $v_i = p$.[1] The first term is behavioral: a higher keep threshold raises return probabilities among those who still buy. The second is selection: a higher price drops cutoff types at the purchase margin, and that term would be negative if those cutoff types return more than the average purchaser.[2] Behavioral and selection effects can therefore pull in opposite directions, and we do not impose a sign on $r'(p)$.

The product market is perfectly competitive. Producer j takes p as given, chooses output at cost $C_j(q_j)$, and incurs a net return handling cost h on each returned unit, equal to gross handling cost minus the scrap value of the returned product. This net cost may be negative, but we maintain $p + h > 0$, which holds whenever scrap value does not exceed list price.

3 Consumer side

We turn first to consumer surplus and show that only the aggregates D and r enter. Write $cs(v,p)$ for expected surplus at list price p among types with willingness to pay v, so $cs(v,v) = 0$. Consumer surplus integrates surplus over all types who buy,

$$CS(p) = \int_p^\infty cs(v,p)\, dF(v). \tag{1}$$

Differentiating in p gives an extensive-margin term from types entering or leaving the purchase margin and an intensive-margin term from lower surplus among inframarginal buyers,

$$CS'(p) = -cs(p,p)\, f(p) + \int_p^\infty cs_p(v,p)\, dF(v),$$

where $cs_p(v,p) \equiv \partial cs(v,p)/\partial p$. On the extensive margin, $cs(p,p) = 0$, so entry and exit contribute nothing at first order. On the intensive margin, $cs_p(v,p)$ equals minus the keep probability, because a higher price harms the consumer only when she keeps the purchase,

$$cs_p(v,p) = -\Pr(u \geq p - \kappa_i).$$

Aggregating over buyers who purchase then gives

$$CS'(p) = \int_p^\infty cs_p(v,p)\, dF(v) = -D(p)\big(1 - r(p)\big),$$

by the definitions of $D(p)$ and $r(p)$. A small price change therefore satisfies

$$dCS = -D(p)\big(1 - r(p)\big)\, dp. \tag{2}$$

[1] With $\rho(v,p)$ as defined in the text, $r(p) = D(p)^{-1}\int_p^\infty \rho(v,p)\, dF(v)$. Differentiating and using $D'(p) = -f(p)$ gives

$$r'(p) = D(p)^{-1}\int_p^\infty \partial\rho(v,p)/\partial p\, dF(v) + \frac{f(p)}{D(p)}\big(r(p) - \rho(p,p)\big),$$

which is the display in the text.

[2] Selection need not be negative for arbitrary (κ_i, G_i). It is signed under an additive match specification $u = v_i + \varepsilon$ with shock ε of any distribution independent of willingness to pay, and with return hassle likewise independent of v: then $\rho(v,p)$ decreases in v, so $\rho(p,p) \geq r(p)$ and the selection term is non-positive.

The extensive- and intensive-margin decomposition makes the local formula transparent. Consumer surplus changes depend only on the aggregates $D(p)$ and $r(p)$, both at each price and integrated along any path,

$$\Delta CS = -\int_{p_0}^{p_1} D(p)\big(1 - r(p)\big)\, dp.$$

Micro heterogeneity in price responses can be rich. Price p enters each $cs_i(p)$ nonlinearly and can shift participation, return decisions, and inframarginal surplus differently across types. Yet once (D, r) are given, this microstructure is irrelevant for dCS and ΔCS. Two economies with the same paths of $D(p)$ and $r(p)$ deliver the same consumer surplus loss, whether high-v_i types adjust return behavior more steeply than low-v_i types or not. Only kept units enter at first order, through $1 - r(p)$, not gross purchases $D(p)$ alone.

Let $\bar{p}$ denote the highest willingness to pay in the population, so $D(\bar{p}) = 0$ and $CS(\bar{p}) = 0$. Integrating the local formula then rewrites consumer surplus as area under kept volume,

$$CS(p) = \int_{p}^{\bar{p}} D(z)\big(1 - r(z)\big)\, dz. \tag{3}$$

4 Supply side

On the firm side, returns make list price an imperfect summary of producer payoffs. A gross sale brings revenue p if the consumer keeps the item and costs refund p plus net handling h if the item is returned. Among gross sales at list price p, share $r(p)$ are returned, so expected revenue per gross unit—the *effective marginal revenue*—is

$$\mu(p, h) \equiv p - r(p)(p + h). \tag{4}$$

Under perfect competition, firm j takes p as given and therefore faces a fixed number μ in its supply decision. At equilibrium, $\mu = \mu(p, h)$.

Given μ, firm j chooses output to solve

$$\max_{q_j \geq 0} \; \mu\, q_j - C_j(q_j), \qquad \text{FOC: } C_j'(q_j) = \mu \;\; (\text{if } q_j > 0),$$

with $C_j(0) = 0$. Write $q_j(\mu)$ for firm j's supply, set to zero when μ is below its minimum marginal cost. We assume each C_j has strictly increasing marginal cost, so $q_j'(\mu) > 0$ for active firms.

Summing across firms defines market supply and industry profit,

$$S(\mu) \equiv \sum_j q_j(\mu), \qquad PS(\mu) \equiv \sum_j \big[\mu\, q_j(\mu) - C_j\big(q_j(\mu)\big)\big]. \tag{5}$$

Under that assumption, $S'(\mu) > 0$. Market supply therefore rises with effective revenue, but not necessarily with list price. Firms care about μ, and $\mu(p, h)$ can fail to rise with p when refunds and handling eat into the list-price gain. Which firms are active at each μ determines both the market quantity and total profit.

Differentiating $PS(\mu)$ with respect to μ gives, for each firm,

$$\frac{d}{d\mu}\Big[\mu\, q_j - C_j(q_j)\Big] = q_j + q_j'(\mu)\,\big[\mu - C_j'(q_j)\big].$$

Active firms satisfy $C'_j(q_j) = \mu$ at their chosen output, so the adjustment term $q'_j(\mu)\,[\mu - C'_j(q_j)]$ is zero for each firm—equivalently, the envelope theorem applied to each firm's profit problem gives $d\pi_j/d\mu = q_j(\mu)$. Summing over firms,

$$PS'(\mu) = \sum_j q_j(\mu) = S(\mu).$$

With no production at $\mu = 0$, it follows that industry profit equals the area to the left of the market supply curve,

$$PS(\mu) = \int_0^{\mu} S(z)\,dz. \tag{6}$$

At equilibrium $\mu = \mu(p,h)$, $PS = \int_0^{\mu(p,h)} S(z)\,dz$.

5 Failure of tax neutrality

With firms responding to μ rather than list price, a natural question is whether remittance of a unit tax remains neutral. Weyl and Fabinger (2013) show that without returns, buyer and seller remittance deliver the same equilibrium. With returns, we ask whether that equivalence survives by writing the two clearing conditions at a common checkout price p—the total payment at the register—and a per-unit tax τ. Fix p and τ. Seller remittance clears at

$$D(p) = S\big(p - \tau - r(p)(p+h)\big), \tag{7}$$

because both the return rate and per-return cost $p + h$ are evaluated at checkout price p. Buyer remittance clears at

$$D(p) = S\big((p-\tau) - r(p-\tau)(p-\tau+h)\big), \tag{8}$$

because the seller posts list price $p - \tau$, so returns and refund cost are evaluated there. The two conditions define the same equilibrium only if the arguments of $S(\cdot)$ coincide. Canceling $p-\tau$ yields

$$r(p)(p+h) = r(p-\tau)(p-\tau+h). \tag{9}$$

Result 1 (Failure of tax neutrality with returns)**.** *Seller and buyer taxes define the same equilibrium if and only if* $r(p)(p+h) = r(p-\tau)(p-\tau+h)$.[3]

Tax neutrality breaks down because the refunded amount—and therefore the keep-or-return decision—depends on which party remits the tax. Under seller remittance the tax-inclusive checkout price is refunded; under buyer remittance only the pre-tax list price is. Purchase is always based on checkout price p, but returns are evaluated at p under seller remittance and at $p-\tau$ under buyer remittance, so effective revenue generally differs and the two clearing conditions are not equivalent.

6 Pass-through

Equilibrium satisfies $D(p) = S(\mu(p,h))$. Write $Q \equiv D(p)$. We characterize pass-through into list price for three shocks: a uniform one-dollar parallel shift in every firm's marginal production cost (indexed by c), a one-dollar increase in handling cost h, and a unit tax under seller and buyer

[3] Neutrality requires the return wedge to be the same at checkout price p and at posted list price $p-\tau$. That holds for all taxes only if $r(p)(p+h)$ is constant in posted list price, which requires $r(p) = C/(p+h)$ for some constant C; without that functional form on r, the two clearing conditions need not define the same equilibrium.

remittance. Production cost shifts market supply at a given effective margin μ, while handling lowers $\mu(p, h)$ at a fixed list price p.

For production and handling, write the clearing condition as

$$m(p, x) \equiv D(p) - S(\mu(p, x), x) = 0, \qquad x \in \{c, h\}. \tag{10}$$

Differentiating while keeping $m = 0$ defines pass-through $\rho_x \equiv dp/dx$ by

$$\rho_x = -\frac{m_x}{m_p}, \tag{11}$$

and expanding $m = D - S$ gives

$$m_p = D'(p) - S'(\mu)\,\mu_p, \qquad m_x = -S'(\mu)\,\mu_x - S_x, \tag{12}$$

hence

$$\rho_x = \frac{-S'(\mu)\,\mu_x - S_x}{S'(\mu)\,\mu_p - D'(p)}. \tag{13}$$

The two shocks differ in which margin moves at fixed p or fixed μ,

$$\mu_c = 0, \qquad S_c = -S'(\mu), \qquad \mu_h = -r(p), \qquad S_h = 0. \tag{14}$$

The production term is $S_c = -S'(\mu)$ because a one-dollar parallel rise in every firm's marginal cost is equivalent, at fixed μ, to evaluating market supply at $\mu - 1$. Handling moves μ by only $r(p)$ per dollar relative to that unit production shift, so its numerator in (13) is the fraction $r(p)$ of the production numerator.

Turning to the unit tax, failure of tax neutrality requires separate checkout prices p^s and p^b under seller and buyer remittance. Write $\rho^s_\tau \equiv dp^s/d\tau$ and $\rho^b_\tau \equiv dp^b/d\tau$. At $\tau = 0$, (13) with $S_\tau = 0$ gives

$$\rho^s_\tau = \frac{-S'(\mu)\,\mu^s_\tau}{S'(\mu)\,\mu_p - D'(p^s)}, \qquad \rho^b_\tau = \frac{-S'(\mu)\,\mu^b_\tau}{S'(\mu)\,\mu_p - D'(p^b)}. \tag{15}$$

The regimes differ only through μ^s_τ and μ^b_τ at a fixed checkout price. At $\tau = 0$, $\mu^s_\tau = -1$. Under buyer remittance, τ enters effective revenue in the same way as $-p$: a tax increase lowers the pre-tax price entering $\mu(p - \tau, h)$ by one dollar, changing μ by $-\mu_p$. Hence $\mu^b_\tau = -\mu_p$ and $\rho^b_\tau = \mu_p\,\rho_c$. The sign of buyer-remitted pass-through is thus tied to that of seller-remitted pass-through by the sign of μ_p. When returns are absent ($r = 0$), $\mu_p = 1$ and the two pass-through rates coincide, recovering the classical neutrality benchmark. If $\mu_p < 0$, however, the two rates always carry opposite signs—a striking reversal of that neutrality intuition.

Result 2 (Pass-through into list price and remittance-dependent sign)**.** ***(i)*** *Production-cost pass-through is*

$$\rho_c = \frac{S'(\mu)}{S'(\mu)\,\mu_p - D'(p)}. \tag{16}$$

The remaining shocks connect to this object by

$$\rho_h = r(p)\,\rho_c, \qquad \rho^s_\tau = \rho_c, \qquad \rho^b_\tau = \mu_p\,\rho_c. \tag{17}$$

(ii) *If $\mu_p < 0$, then*

$$\operatorname{sgn}(\rho^b_\tau) = -\operatorname{sgn}(\rho_c)$$

provided $\rho_c \neq 0$. A unit tax therefore moves the equilibrium checkout price in opposite directions depending on which side of the market remits it.

The condition $\mu_p < 0$ requires the behavioural response of returns to price to be strong enough that effective revenue falls when p rises: from $\mu_p = 1 - r(p) - r'(p)(p+h)$, we have $\mu_p < 0$ exactly when $r'(p)(p+h) > 1 - r(p)$. In that case buyer remittance raises μ while seller remittance lowers it, shifting supply outward and inward respectively. The checkout price must therefore adjust in opposite directions to restore equilibrium. Which regime raises the price and which lowers it depends on the sign of ρ_c, which in turn hinges on the relative magnitudes of the supply and demand elasticities—a point developed in the next subsection.

6.1 Elasticity representation

The same pass-through formulas can be rewritten in elasticities, which makes the role of μ_p transparent. Write $\mu_p \equiv d\mu/dp$ for the response of effective revenue to list price, and define demand and supply elasticities with respect to list price as usual by

$$\varepsilon_D(p) \equiv \frac{pD'(p)}{D(p)} < 0, \qquad \varepsilon_S(p) \equiv \frac{dS(\mu(p))}{dp}\,\frac{p}{S(\mu(p))} = S'(\mu)\,\mu_p\,\frac{p}{S(\mu)}.$$

Here $\varepsilon_S(p)$ is the observed elasticity of market supply with respect to the clearing price. Without returns, $\mu_p = 1$ and $\varepsilon_S > 0$. With returns, μ_p can become negative, in which case $\varepsilon_S < 0$ and a higher list price reduces the quantity firms are willing to supply.

Without returns, supply is a function of list price, so $S' = dS/dp$. A one-dollar cost increase cuts supply by S' units at the original price, creating an excess-demand gap of that size. Raising list price by one dollar closes the gap from both sides. Demand falls by $|D'|$ and supply recovers by S'. Closing a gap of size S' therefore requires a price rise of $S'/(S'+|D'|)$, so standard pass-through equals $\varepsilon_S/(-\varepsilon_D + \varepsilon_S)$.

With returns, supply is a function of effective revenue, so $S'(\mu) = dS/d\mu$ rather than dS/dp. A one-dollar production-cost increase is equivalent to lowering μ by one dollar, so at fixed list price supply falls by $S'(\mu)$ units, again creating an excess-demand gap of that size. Raising list price by one dollar closes the gap at rate $S'(\mu)\,\mu_p - D'(p)$, so

$$\rho_c = \frac{S'(\mu)}{S'(\mu)\,\mu_p - D'(p)}.$$

The cost shock lowers μ by one dollar, while a dollar of list price raises μ by μ_p. A higher μ_p means a one-dollar price increase restores more effective revenue—and therefore more quantity supplied—than the one-dollar cost shock removes, so a smaller list-price movement closes the excess-demand gap. Measuring supply against list price through ε_S puts the factor $1/\mu_p$ in the numerator,

$$\rho_c = \frac{\varepsilon_S(p)/\mu_p}{-\varepsilon_D(p) + \varepsilon_S(p)}. \tag{18}$$

When $\mu_p < 0$, the supply curve expressed as a function of the list price p is locally decreasing. In that region the sign of ρ_c is no longer guaranteed positive. It depends on the relative magnitudes of ε_S and ε_D. This opens the door to the counterintuitive outcome that an increase in marginal cost reduces the equilibrium price.

Result 3 (Negative pass-through)**.** *We have $\rho_c < 0$ if and only if $\varepsilon_S < 0$ ($\mu_p < 0$) and $|\varepsilon_S| > -\varepsilon_D$.*

Proof. By (18),

$$\rho_c = \frac{\varepsilon_S/\mu_p}{-\varepsilon_D + \varepsilon_S}.$$

Since $\varepsilon_S = S'(\mu)\mu_p\, p/S(\mu)$, the terms ε_S and μ_p have the same sign, so $\varepsilon_S/\mu_p > 0$. Hence $\rho_c < 0$ if and only if

$$-\varepsilon_D + \varepsilon_S < 0,$$

which is equivalent to $\varepsilon_S < 0$ and $|\varepsilon_S| > -\varepsilon_D$. Because ε_S and μ_p have the same sign, $\varepsilon_S < 0$ is equivalent to $\mu_p < 0$. □

The logic is as follows. With $\varepsilon_S < 0$ ($\mu_p < 0$), a higher list price lowers the quantity firms supply, so a leftward shift of supply—caused by a rise in c—creates excess demand at the initial p. Raising the price to eliminate the excess would further contract supply and worsen the imbalance; instead, the price must fall. A decline in p increases both the quantity demanded and, because supply slopes downward, the quantity supplied. When $|\varepsilon_S| > -\varepsilon_D$, the supply curve is steeper than the demand curve, meaning that the supply response to a price cut is larger than the demand response. The price decline therefore eliminates the excess demand and leads to a new equilibrium at a lower list price. This sign reversal cannot occur in the classical model, where $\mu_p = 1$ and supply is everywhere upward-sloping, highlighting a qualitative failure of the no-return benchmark.

At first glance, it may seem puzzling why the price drop—which raises effective revenue ($\mu_p < 0$ and $\rho_c < 0$)—necessarily more than offsets the original one-dollar cost increase. The key is that the price fall must do double duty to clear the market. A one-dollar cost increase shifts the supply curve left, creating an initial excess demand gap. To fill this gap, quantity supplied must increase. Since quantity supplied increases with effective revenue μ, restoring enough supply to cover the original gap already requires μ to rise by one dollar. But the price fall also increases quantity demanded, creating additional excess demand on top of the initial gap. To satisfy both the original supply shortfall and this new demand, supply must increase by more than the original shift, so effective revenue must rise by more than one dollar. Thus, $\mu_p\rho_c > 1$ is precisely the market-clearing requirement when pass-through is negative.

Even when $\mu_p > 0$ and pass-through stays positive, the returns-adjusted formula differs from the no-return case whenever $\mu_p \neq 1$. Comparing the two signs the bias for an analyst who ignores returns. Whether μ_p exceeds one depends on how steeply the return rate falls with price. Differentiating $\mu = p - r(p)(p+h)$ gives

$$\mu_p = 1 - r(p) - r'(p)(p+h).$$

Write $\varepsilon_r(p) \equiv p\, r'(p)/r(p)$ for the elasticity of the return rate in list price. Then $\mu_p > 1$ if and only if $\varepsilon_r(p) < -p/(p+h)$. That is, selection must outweigh the behavioral rise in individual return probabilities by enough that aggregate r falls steeply in p. When instead the behavioral response dominates, ε_r is less negative (or positive) and $\mu_p < 1$.

Result 4. *Relative to the correct returns-adjusted ρ_c, the classical no-return pass-through formula overestimates pass-through when $\varepsilon_r(p) < -p/(p+h)$ and underestimates it otherwise.*

The practical value is that the bias can be signed without estimating the full returns-adjusted model. The return-rate elasticity and the ratio of handling to list price are enough. The economics behind the threshold is familiar from the selection and behavioral forces already in ε_r. When selection dominates, a higher list price lowers the return rate, so the refund-and-handling drag on effective revenue shrinks as price rises. Effective revenue can then rise by more than one dollar per dollar of list price, and a smaller price movement closes the excess-demand gap than the classical formula predicts. Costly handling strengthens that channel and makes overestimation more likely. Otherwise, effective revenue rises by less than one dollar per dollar of list price, so list price must move by more than the classical formula predicts to close the gap, and the classical no-return formula underestimates pass-through.

7 Local incidence

Pass-through into list price determines the welfare loss from each cost shifter and how that loss is split between consumers and producers. Throughout, $Q \equiv D(p)$ and $r \equiv r(p)$ denote the equilibrium gross quantity and the return rate. From the consumer-side analysis in Section 3, a small increase in any shifter x changes consumer surplus by

$$\frac{dCS}{dx} = -Q(1-r)\rho_x, \tag{19}$$

where $\rho_x \equiv dp/dx$ is the pass-through of the shock into the list price.

Producer surplus is $PS(\mu, x) = \int_0^{\mu} S(z,x)\,dz$. Differentiating with respect to x and applying the equilibrium condition $S(\mu, x) = Q$ gives

$$\frac{dPS}{dx} = Q\frac{d\mu}{dx} + \Sigma_x,$$

where $\Sigma_x \equiv \int_0^{\mu}(\partial S(z,x)/\partial x)\,dz$ captures the direct shift in the area under the supply curve. Expanding the total derivative of the effective margin via the chain rule, $\frac{d\mu}{dx} = \mu_p\rho_x + \mu_x$, yields

$$\frac{dPS}{dx} = Q(\mu_p\rho_x + \mu_x) + \Sigma_x. \tag{20}$$

The local incidence ratio is $I_x \equiv (dCS/dx)/(dPS/dx)$, so

$$I_x = \frac{-Q(1-r)\rho_x}{Q(\mu_p\rho_x + \mu_x) + \Sigma_x}.$$

The components needed to evaluate I_x for each shock are reported in Table 1. Across the four shocks the last column collapses to the same expression in ρ_c and μ_p: the shock-specific pass-through and direct effects cancel in the ratio.

Shock x	ρ_x	μ_x	Σ_x	dCS/dx	dPS/dx	I_x
Production cost c	ρ_c	0	$-Q$	$-Q(1-r)\rho_c$	$Q(\mu_p\rho_c - 1)$	$\frac{(1-r)\rho_c}{1-\mu_p\rho_c}$
Handling cost h	$r\rho_c$	$-r$	0	$-Q(1-r)r\rho_c$	$Qr(\mu_p\rho_c - 1)$	$\frac{(1-r)\rho_c}{1-\mu_p\rho_c}$
Seller tax τ^s	ρ_c	-1	0	$-Q(1-r)\rho_c$	$Q(\mu_p\rho_c - 1)$	$\frac{(1-r)\rho_c}{1-\mu_p\rho_c}$
Buyer tax τ^b	$\mu_p\rho_c$	$-\mu_p$	0	$-Q(1-r)\mu_p\rho_c$	$Q\mu_p(\mu_p\rho_c - 1)$	$\frac{(1-r)\rho_c}{1-\mu_p\rho_c}$

Table 1: Pass-through, direct effects, surplus changes, and incidence ratio for each shock.

Result 5. *For all four shocks, the local incidence ratio is identical:*

$$I_x = \frac{(1-r)\rho_c}{1-\mu_p\rho_c} > 0. \tag{21}$$

The denominator $1-\mu_p\rho_c$ is the per-unit loss in producer surplus. A \$1 cost shock initially reduces producer surplus by Q. The equilibrium list-price rise ρ_c increases effective marginal revenue per unit by $\mu_p\rho_c$. Because $dPS/d\mu = Q$ by the envelope theorem, this marginal gain translates into a total recovery of $Q\mu_p\rho_c$ in producer surplus. The net loss is therefore $Q(1-\mu_p\rho_c)$. The numerator $(1-r)\rho_c$ is the corresponding per-unit consumer loss on kept units. The incidence ratio compares these two per-unit burdens. A small denominator leaves consumers bearing almost the entire shock, while a negative denominator, $\mu_p\rho_c > 1$, means the recovery exceeds the cost, so producers are net gainers. This recovery factor can be written as $\mu_p\rho_c = \varepsilon^S/(\varepsilon^S - \varepsilon^D)$. This expression closely mirrors the pass-through formula, and now we argue that $\rho_c < 0$ if and only if $\mu_p\rho_c > 1$. The condition $\mu_p\rho_c > 1$ requires $\varepsilon^S < 0$, which implies $\mu_p < 0$, and hence $\rho_c < 0$. Conversely, $\rho_c < 0$ requires $\varepsilon^S < 0$ and $|\varepsilon^S| > -\varepsilon^D$, which gives $\mu_p\rho_c > 1$. Therefore, the condition governing producer gain from an increase in production cost is exactly the same as that governing negative pass-through.

At first glance, it may seem puzzling why the price drop—which raises effective revenue ($\mu_p < 0$ and $\rho_c < 0$)—necessarily more than offsets the original one-dollar cost increase. The key is that the price fall must do double duty to clear the market. A one-dollar cost increase shifts the supply curve left, creating an initial excess demand gap. To fill this gap, quantity supplied must increase. Since quantity supplied increases with effective revenue μ, restoring enough supply to cover the original gap already requires μ to rise by one dollar. But the price fall also increases quantity demanded, creating additional excess demand on top of the initial gap. To satisfy both the original supply shortfall and this new demand, supply must increase by more than the original shift, so effective revenue must rise by more than one dollar. Thus, $\mu_p\rho_c > 1$ is precisely the market-clearing requirement when pass-through is negative.

The preceding observation implies a striking result: the uniformly positive incidence ratio implies that consumer and producer surplus always move together, regardless of the sign of pass-through. The direction of that common movement depends on both the shock type and the sign of pass-through. When $\rho_c < 0$, production cost, handling cost, and seller tax shocks generate a welfare gain with both surpluses increasing. Under the same condition, buyer-remitted taxes reduce both surpluses. The two tax regimes therefore yield opposite welfare outcomes precisely when pass-through is negative, a sharp violation of classical tax neutrality. When $\rho_c > 0$, all four shocks reduce both surpluses. The central asymmetry is twofold. First, seller and buyer tax remittance can lead to divergent welfare consequences. Second, welfare-improving shocks are possible only for a subset of shocks (cost, handling, seller tax) when pass-through is negative. Neither possibility arises in the no-return benchmark, where pass-through is always positive.

7.1 Deadweight loss and the returns-augmented Harberger triangle

The marginal change in private surplus $W = CS + PS$ from a small shock x follows by summing (19) and (20) and substituting $\mu_p = 1 - r - r'(p)(p+h)$,

$$\frac{dW}{dx} = Q\left[\mu_x - r'(p)(p+h)\rho_x\right] + \Sigma_x, \tag{22}$$

with μ_x and Σ_x as in Table 1. The term $-Q\,r'(p)(p+h)\rho_x$ is new relative to the classical model. It captures the efficiency consequence of the price-induced change in the return rate. Each returned unit consumes real handling resources h, so a change in the return rate has a first-order welfare impact that is absent when returns are shut down.

Without returns, $r = 0$ and $r' = 0$, so (22) reduces to

$$\frac{dW}{dx} = Q\,\mu_x + \Sigma_x.$$

For a unit production-cost increase, $\mu_c = 0$ and $\Sigma_c = -Q$, giving $dW/dc = -Q$: a pure rectangle of lost surplus, as society must devote an additional Q units of resources to produce the same output.

For a unit tax τ with tax revenue excluded from W, one likewise obtains $dW/d\tau = -Q$, the transfer from private agents to the government. Adding tax revenue back, total surplus $V = W + \tau Q$ has a zero first-order change at $\tau = 0$. The inefficiency of the tax appears only at second order, giving the classic Harberger triangle $\Delta V \approx -\frac{1}{2}\tau^2 Q'(0)$. Cost shocks therefore generate a first-order rectangle and taxes a second-order triangle. The two efficiency concepts are cleanly separated.

Product returns destroy this separation. The return-rate term in (22) is generally nonzero for every shock, so cost, handling, and both remittance regimes of a unit tax all carry a first-order welfare change that is not a simple transfer. Using Table 1 and the pass-through links in Result 2, write

$$\Delta \equiv 1 + r'(p)(p+h)\,\rho_c.$$

Then

$$\frac{dW}{dc} = -Q\,\Delta, \qquad \frac{dW}{dh} = -Q\,r\,\Delta, \qquad \frac{dW}{d\tau^s} = -Q\,\Delta, \qquad \frac{dW}{d\tau^b} = -Q\,\mu_p\,\Delta. \tag{23}$$

When $\Delta < 0$, which coincides with negative pass-through $\rho_c < 0$, a rise in production cost, handling cost, or a seller-remitted tax increases private surplus—a first-order Pareto improvement that is impossible in the no-return benchmark. A buyer-remitted tax is scaled by μ_p. If additionally $\mu_p < 0$, its first-order effect has the opposite sign of the seller-tax effect.

The displays in (23) are local. The global welfare change from a discrete shock Δx integrates from the initial equilibrium to Δx,

$$\Delta W = \int_0^{\Delta x} \frac{dW}{dx}\,dx.$$

Expanding the integrand around $x = 0$, the first-order term is the marginal loss at zero times the shock size, and the second-order term is the returns-augmented Harberger triangle,

$$\Delta W \approx \left.\frac{dW}{dx}\right|_{x=0} \Delta x + \text{second-order triangle}. \tag{24}$$

Let $\varepsilon_D = pD'(p)/Q$, and write $\varepsilon_\mu = \mu\,S'(\mu)/Q$ for the elasticity of supply with respect to effective revenue. The observed list-price supply elasticity is then $\varepsilon_S = \varepsilon_\mu\,\mu_p\,(p/\mu)$, which reduces to the ordinary supply elasticity when $\mu_p = 1$. With returns, μ_p may be negative, so ε_S may be negative and the triangle can change sign.

For a production-cost shock Δc (a parallel shift of effective supply),

$$\text{Triangle}_c \approx \tfrac{1}{2}(\Delta c)^2\,\frac{-\varepsilon_D\,\varepsilon_S}{\varepsilon_S - \varepsilon_D}\,\frac{Q}{p}. \tag{25}$$

A handling-cost increase Δh shifts effective supply by $r\,\Delta h$, so

$$\text{Triangle}_h \approx \tfrac{1}{2}(r\,\Delta h)^2\,\frac{-\varepsilon_D\,\varepsilon_S}{\varepsilon_S - \varepsilon_D}\,\frac{Q}{p} = r^2\,\text{Triangle}_c\big|_{\Delta c = \Delta h}. \tag{26}$$

A seller-remitted tax τ^s is identical to a production-cost increase in μ-space,

$$\text{Triangle}_{\tau^s} \approx \tfrac{1}{2}(\tau^s)^2\,\frac{-\varepsilon_D\,\varepsilon_S}{\varepsilon_S - \varepsilon_D}\,\frac{Q}{p}. \tag{27}$$

A buyer-remitted tax τ^b wedges the buyer price against the pre-tax price that enters effective revenue, with pass-through $\rho^b_\tau = \mu_p \rho_c$. Its triangle is

$$\text{Triangle}_{\tau^b} \approx \tfrac{1}{2}(\tau^b)^2\,\frac{-\varepsilon_D\,(\varepsilon_S/\mu_p)}{\varepsilon_S/\mu_p - \varepsilon_D}\,\frac{Q}{p} = \tfrac{1}{2}(\tau^b)^2\,\frac{-\varepsilon_D\,\varepsilon_\mu\,(p/\mu)}{\varepsilon_\mu\,(p/\mu) - \varepsilon_D}\,\frac{Q}{p}, \tag{28}$$

using $\varepsilon_S/\mu_p = \varepsilon_\mu\,(p/\mu)$. If $\mu_p < 0$, this triangle has the opposite sign of the seller-tax triangle.

Combining the first-order rectangles in (23) with these second-order triangles gives the efficiency cost of each shock. Unlike the classical model, the rectangle can dominate the triangle even for small shocks, and both components can change sign with the return-rate response. When $\Delta < 0$, the first-order gain from a cost increase or seller tax may be large enough that private surplus rises even after the (possibly negative) triangle—a stark reversal of standard intuition.